Uncanny Valley

Article Summaries — Section 1

Mori 1970 • MacDorman & Ishiguro 2006 • Ho & MacDorman 2010

McDonnell et al. 2012 • Dill et al. 2012 • Seymour et al. 2021

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Mori (1970) — The Uncanny Valley

Mori (1970)

Foundational theoretical paper — proposes the human affinity curve for robots and artificial figures

Key Findings

The valley: affinity rises with human-likeness, then drops sharply at the near-human threshold, recovering only at complete fidelity

The problem zone: modern production avatars sit at mid-range realism — exactly where the drop is deepest and discomfort is strongest

Movement amplifies: a moving near-human triggers stronger aversion than a static one at the same realism level — critical for animated avatars

What this study adopts

Core motivation for NPR: stylisation deliberately places the avatar below the uncanny zone, trading photorealistic fidelity for affective safety. Directly motivates Research Questions 1 and 3.

MacDorman & Ishiguro (2006) — Uncanny Valley Empirical Programme

MacDorman & Ishiguro (2006)

Proposes semantic-differential scales to operationalise eeriness as a quantifiable experimental variable

Key Findings

Category mismatch theory: the UVE is driven not by imperfection in appearance, but by categorical ambiguity — agents positioned between “human” and “non-human” categories produce a cognitive mismatch and a negative affective response

Semantic-differential scales: proposed measurement instruments for eeriness and strangeness, converting Mori’s qualitative curve into a quantifiable experimental variable used in subsequent validation work

Androids as tools: androids can be positioned at any point along the human-likeness continuum with experimental control, enabling controlled generation of UVE responses on demand

What this study adopts

Original source for the eeriness measurement approach in Part D. The category mismatch account supports the prediction that NPR stylisation reduces eeriness by shifting the Avaturn avatar away from the ambiguous near-human boundary, even without dramatically reducing perceived human-likeness.

Ho & MacDorman (2010) — Eeriness Scale

Ho & MacDorman (2010)

Validated alternative to Godspeed for uncanny valley research; separates eeriness from human-likeness

Key Findings

The Godspeed problem: existing eeriness items conflated discomfort with human-likeness — two dimensions that can vary independently

Key distinction: an avatar can look human without feeling eerie, or feel eerie while appearing non-human — separate perceptual dimensions

The instrument: 6 semantic-differential items, 5-point, two subscales (eeriness and human-likeness) validated across robotic and virtual stimuli

What this study adopts

Part D — 6 eeriness items, 5-point (adapted). Directly tests H1: does NPR reduce eeriness? Eeriness and human-likeness are measured separately to detect if NPR reduces uncanniness without reducing perceived human-likeness.

McDonnell et al. (2012) — Render Me Real?

McDonnell et al. (2012)

11 rendering styles (ToonPencil to photorealistic); trust, appeal, eeriness, and lie-detection across still and moving stimuli

Key Findings

Non-linear trustworthiness: both the most abstract (toon) and most photorealistic styles received equivalent trust ratings — mid-range styles scored lowest. Directly replicates the valley shape for trustworthiness

Stylised can exceed photorealistic on comfort: ToonFlat and ToonShaded were rated more reassuring than near-photorealistic HumanHQ2 — highly stylised rendering can surpass photorealistic rendering on perceived comfort

Motion amplifies the valley: HumanIII became significantly less appealing when moving ($p < .009$) — imperfect motion on a near-realistic avatar is perceptually more costly than on a stylised one, supporting the video stimulus choice

What this study adopts

Direct empirical precedent for H2 (NPR does not reduce trust relative to photorealistic rendering). Finding that abstract and photorealistic styles receive statistically equivalent trustworthiness is the clearest published evidence. Also supports the choice of video stimuli: motion amplifies the valley for mid-range unappealing styles.

Dill et al. (2012) — Evaluation of the Uncanny Valley in CG Characters

Dill et al. (2012)

10 CG characters from recent movies and games; two-stage questionnaire (still images and video); human-likeness vs familiarity ratings

Key Findings

Valley confirmed for CG characters: Mori's familiarity curve was replicated for contemporary movie and game CG characters in both still images and video stimuli

Motion deepens the valley: the uncanny response was stronger for animated than still-image characters, consistent with Mori's original motion hypothesis

Eyes and mouth cause most discomfort: the primary sources of unease were the two most expressive and detail-dependent facial regions

What this thesis cites

Provides empirical evidence that the uncanny valley applies to CG avatars, directly motivating the NPR approach. The eyes-and-mouth finding supports NPR stylisation as a targeted intervention: abstracting the highest-detail facial features reduces the specific discomfort these regions create. The motion amplification result justifies the video-based study design. Supports H1.

Seymour et al. (2021) — Have We Crossed the Uncanny Valley?

Seymour et al. (2021)

Human-realistic avatar (Digital MIKE) vs caricature avatars; VR headset (n=72) and 2D screen (n≈550)

Key Findings

Human-realistic preferred: Digital MIKE received significantly higher affinity, trustworthiness, and preference ratings than caricature avatars. The second peak is now reachable in real-time rendering

Caricatures = positive engagement: caricature avatars produced no overt negative reactions; participants engaged socially with them — consistent with Mori's first slope where low-realism figures produce positive affect

VR amplifies differences: VR participants showed stronger preference for the human-realistic avatar than 2D screen observers, suggesting immersive display heightens perceptual differences between realism levels

What this study adopts

Grounds H3: both avatar conditions (C2, C3) are expected to produce lower trust than the real person (C1). The C2 vs C3 comparison mirrors Seymour's human-realistic vs caricature comparison, though in this study the NPR shader is applied to the same base avatar rather than an entirely distinct caricature character.